

Test Report
Eddy Current Inspection
Project No.: 16.6191.135
QSGTL
2E-3012



Eddy current inspection Test report QSGTL 2E-3012

Page 2 of 7

Date: 02-09-2016

1. General

Realization: ec-works GmbH
Inspection personal: F. Müller, A. Hullmann
Place of inspection: QSGTL, RLIC Qatar
Test object: 2E-3012

System	Outside-diameter	Wall-thickness	Length	Material	No. of legs	Tested legs
2E-3012	19,05	2,11	7.400	Carbon Steel	1140	1138

Tube dimensions in mm

2. Task

With the use of the non-destructive eddy current inspection method in the differential and absolute mode wall thickness decreases had to be detected. The detected inner and outer damages had to be documented with a percentage fault depth compared to the calibration tube sets and with specifications of the position in longitudinal tube axis.

3. Referenced Documents:

EN ISO 9712:2012: Non destructive testing- Qualification and certification of NDT personnel

EN ISO 15549:2010: Non destructive testing – Eddy Current Testing- General Principals

EN ISO 12718:2008: Non destructive testing – Eddy Current Testing - Terms

DIN 54140, Part 3: Non-destructive testing, electromagnetic methods; representation and general characteristics of coil systems



Eddy current inspection Test report QSGTL 2E-3012

Page 3 of 7
Date: 02-09-2016

4. Scope of testing

100% of the tubes

5. Personal:

The system-operator for this eddy current inspection has to be qualified regarding EN 473 or EN ISO 9712. This operator will execute the test including signal interpretation.

The operator has to be skilled with sufficient knowledge of interpretation as well as he has to be familiar with the forms of appearance of defects caused operationally. The evaluation of abnormalities and indications exceeding the result limits has to be carried out by the inspection supervisor. This supervisor needs to be provided with the required experience relative to the inspection task, the inspection method and the setting of devices.

The operator needs to have a good command of the work that has to be performed. Particularly he has to prove satisfactory experience in carrying out eddy current inspections and knowledge about the inspection task in this regard.

The inspection personal have to be certified as shown in table 1.

Personal	Certification
Supervisor of inspection	Level 3 of the appropriate product or industry sector under the terms of EN 473 / ISO 9712
Operator	Level 2 of the appropriate product or industry sector under the terms of EN 473 / ISO 9712

6. Test procedure and test system

6.1 Eddy current test method

Eddy current multi-frequency method using inside passing probes with magnetizing coils (PSEC). The eddy current signals are displayed on the impedance plane.

6.2 Test System:

The used test system consists of:

Test device: Multi-frequency eddy current device

Test probes: Bobbin coils for bridge circuit in difference mode

	Eddy current inspection Test report QSGTL 2E-3012	Page 4 of 7 Date: 02-09-2016
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The cable length including the used probe cable extension is 35 m.

Inspection and result documentation system: „ec-inspector“

7. Test frequency:

With regard to the tube material, tube dimension, a sufficient ability of proving defects as well as an optimal angle of phase displacement for inside and outside defects, the test frequency was set to 90 kHz.

8. Reference tube set:

Regarding the geometry as well as the electric and magnetic characteristics the reference tube set corresponds to the test object. Relating to the defect geometry, artificial defects introduced into the reference tube set should be similar to the defects to be expected and defined in the test object. During the inspection the artificial defects serve as adjustment defects for the device adjustment as well as comparable defects.

Tube set no.: A 081 with simulated corrosion and erosion defects.

Calibration defect: outer pitting 75% wall loss.

9. Test execution:

9.1 Test requirements:

The tubes had to be free of deposits on the inside surface as the probes can be processed without any impediment over the total test area.

9.2 Test speed:

The test will be performed by pulling back the probe manually. With regard to the used low-pass filter the maximal test speed is limited to 40 m/min. Lower test speeds have no influence on the test result.

9.3 Analysis technique:

The eddy current signals were shown and assessed at the impedance plane.

	Eddy current inspection Test report QSGTL 2E-3012	Page 5 of 7 Date: 02-09-2016
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In order to differentiate defects proceeding from the tube inside or outside surface as well as to determine the defect depth, the phase angle and the amplitude of the signal were used, comparing signals delivered by probe with the signals from the reference object. The documentation is done in percentage wall loss..

9.4 Registration limit:

All defect signals permitting a clear interpretation (i. e. useful signal to interference signal ratio greater than or equal 2:1) are to be registered.

9.5 Result limits:

ET-signals with a registered defect depth of 20% wall loss had to be documented.

10. Adjustment and calibration of device:

The calibration of the test system takes place by moving the probe through the test tube.

The test system has to be adjusted including the probe cable length used during the test. This has to be done for each probe used for the inspection and therefore after any exchanges in the measurement chain, e.g. probes, cables or EC-units.

10.1 Adjustment of sensitiveness:

Difference mode

The signal of the outside defect (drilled hole, depth 75%) has to achieve a height of 7 scale parts on the screen. The signal phase has to be set vertical.

Absolute mode

The signal of the outside defect (general wall loss, depth 40%) has to achieve a size of 5 scale parts on the screen. The signal phase has to be set horizontal..

10.2 Control of adjustment:

To ensure the quality of the EC-inspection, control adjustments have to be realized. For the control of phase and sensitiveness, the simulated pitting defects are used. The phase angles of the signals may not vary by more than +/- 5 %. In case of exceeding this value, an adjustment must be done again.

The control adjustment is carried out before the beginning and after the termination of measurement, at the beginning and the end of each shift and after any exchanges in the measurement chain.

	Eddy current inspection Test report QSGTL 2E-3012	Page 6 of 7 Date: 02-09-2016
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10. Inspection results

The tubes had been inspected as described above.
External defects, comparable to pitting corrosion, up to 70% wall loss were found.

Test object	Results
2E-3012	- 1140 legs have been inspected and documented. - 943 legs showed no indications above registration limit. - 119 legs had indications with eddy current signals which are evaluated as <i>outside</i> defects up to class (code) 2. - 34 legs had indications with eddy current signals which are evaluated as <i>outside</i> defects up to class (code) 3. - 22 legs had indications with eddy current signals which are evaluated as <i>outside</i> defects up to class (code) 4. - 9 legs had indications with eddy current signals which are evaluated as <i>outside</i> defects up to class (code) 5. - 4 legs had indications with eddy current signals which are evaluated as <i>outside</i> defects up to class (code) 6. - 6 legs had indications with eddy current signals which are evaluated as <i>outside</i> defects up to class (code) 7. - 1 leg could not be tested because of big dents or deposits. It is marked with "R" (restricted). - 2 legs had already been plugged. They are marked with "P" (plugged) For detailed information please refer to the attached pages of the inspection report.

11. Discussion of results

The common defect is external pitting corrosion.

There is slight corrosion all over the bundle.
Major defects are concentrated in the areas of the inlet nozzles.
Especially inlet 2 is affected.

	Eddy current inspection Test report QSGTL 2E-3012	Page 7 of 7 Date: 02-09-2016
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12. Documentation

The absolute figure and the percentage of the eddy current inspection results are calculated in the inspection summary.

The inspection results are visualized with a defect picture which shows the tube sheet with the deepest indication of the eddy current inspection.

Inspected tubes without any indication above the registration level are marked with "X", no defect

The evaluated defect depth and the axial position are listed at the defect report.

All documentation data have been stored digitally.

The inspection documentation is part of this report with 7 pages.

There are no further attachments to this inspection report.

Doha; 02nd September 2016

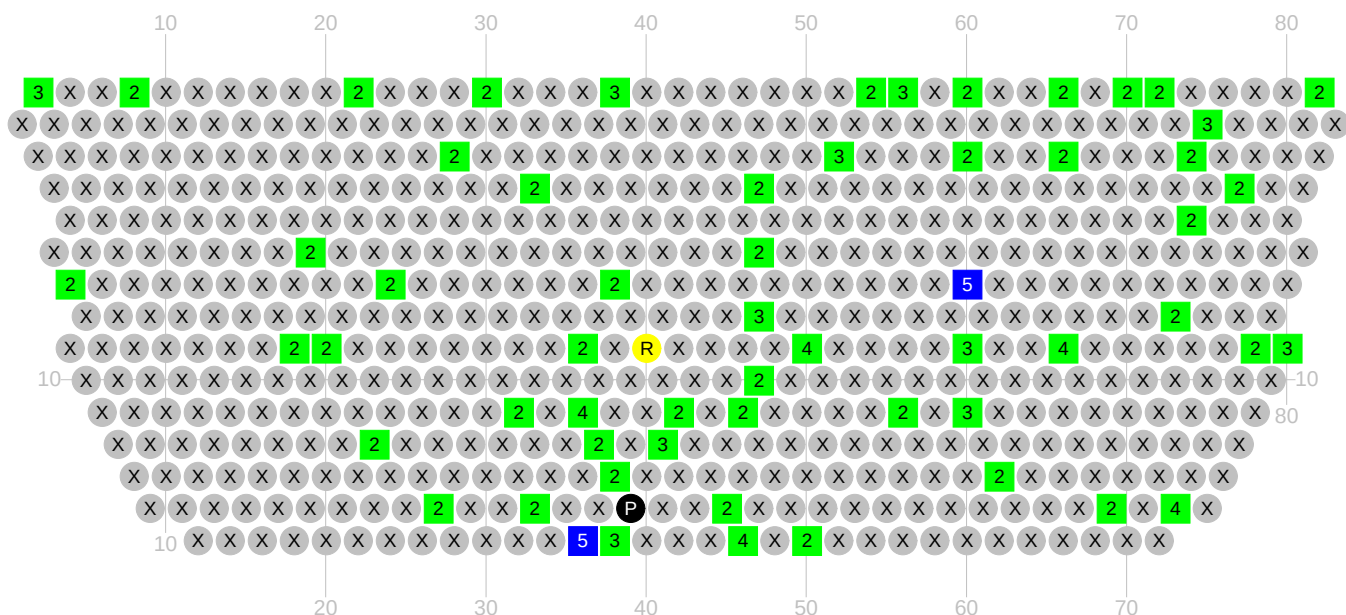
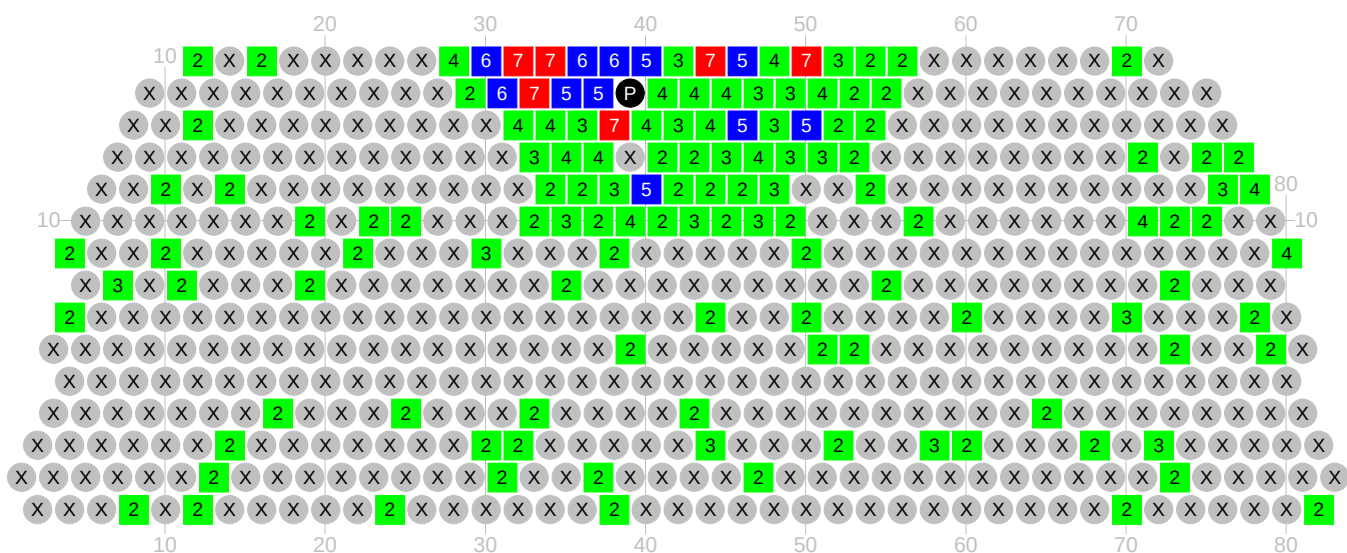


Frank Müller
EN 473 ET Level III

Summary of Inspection Results

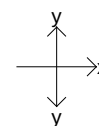
Customer: QSGTL Place: Qatar, Ras Laffan PR-No: 16.6191.135		Date: 24.11.-4713 Object: 2E-3012	
<u>Tube Data</u> Tube Type: U-Tube Material: CS Length: 7.40 m ø outside: 19.05 mm ø inside: 14.83 mm Wall Thickn.: 2.11 mm		<u>Inspection Data</u> Operator: F.Müller/A.Hullmann EC-unit Type: ecw 06 Probe Type: D-B-ID Calib. Tube: A 081 Calib. Defect: pitting 50% out Frequency: 90.00 kHz	
<u>Remarks</u> inspection of straigth section			
Defect Depth	Code	Qty inside	Code Qty outside
10% - 19%	1	0 = 0.00%	1 0 = 0.00%
20% - 29%	2	0 = 0.00%	2 119 = 10.44%
30% - 39%	3	0 = 0.00%	3 34 = 2.98%
40% - 49%	4	0 = 0.00%	4 22 = 1.93%
50% - 59%	5	0 = 0.00%	5 9 = 0.79%
60% - 69%	6	0 = 0.00%	6 4 = 0.35%
70% - 79%	7	0 = 0.00%	7 6 = 0.53%
80% - 89%	8	0 = 0.00%	8 0 = 0.00%
90% - 99%	9	0 = 0.00%	9 0 = 0.00%
Dent	D	0 = 0.00%	
Without Defect	X	943 = 82.72%	
Restricted	R	1 = 0.09%	
Number of Legs		1140 = 100,00%	
Inspected Legs		1138 = 99.82%	
Plugged	P	2 = 0.18%	
To Be Plugged		19 = 1.67%	
Final Number of Plugs		21 = 1.84%	
Available		1119 = 98.16%	
created on 01.09.2016			

Top 2E-3012



Bottom 2E-3012

ec-works



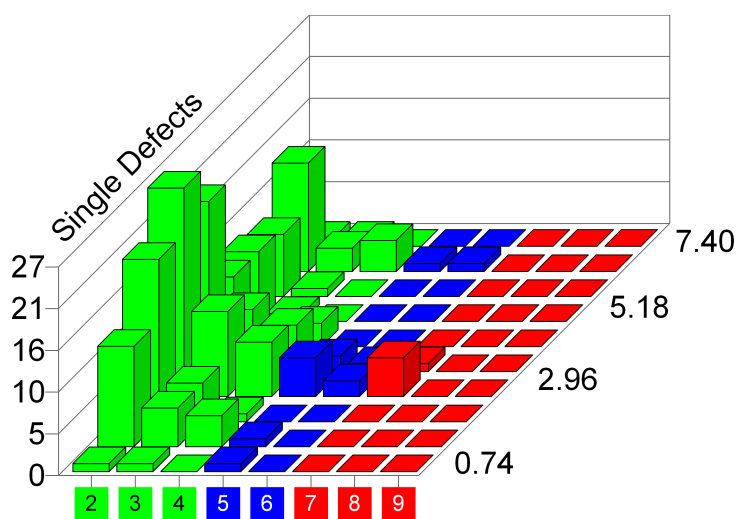
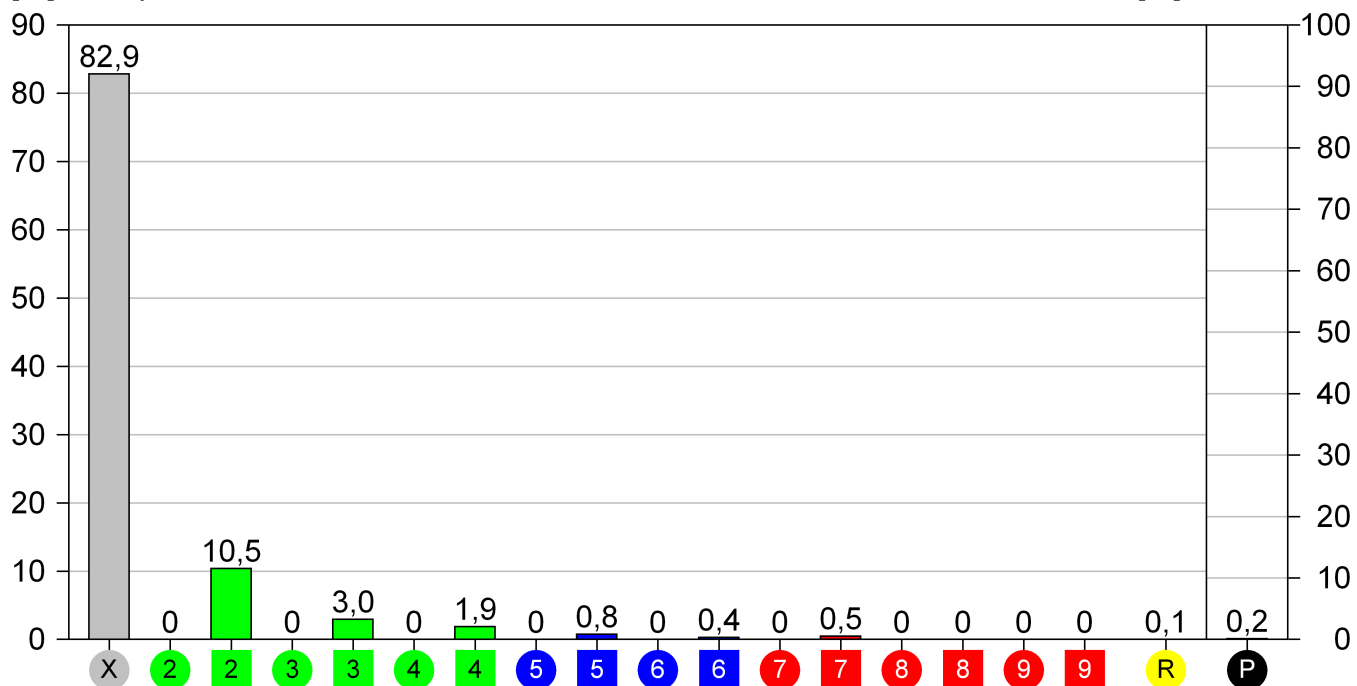
Customer: QSGTL
Object: 2E-3012

Pr-Nr: 16.6191.135
Date: 01.01.1970
View: In/Out

2 - 9 Outside Defects
P Plugged
R Restricted
X Without Defect

[%] of inspected Tubes

[%] of all Tubes



Frequency
Position [m]
Defect Class

Customer: QSGTL
Place: Qatar, Ras Laffan
PR-No: 16.6191.135

Date: 24.11.-4713
Object: 2E-3012
Bundle: Top 2E-3012



Tube coordinates		biggest inner defect		biggest outer defect		biggest other defect		total def.	Tube coordinates		biggest inner defect		biggest outer defect		biggest other defect		total def.
X	Y	%	cm	%	cm	Symb.	cm		X	Y	%	cm	%	cm	Symb.	cm	
8	1			20	652			2	22	9			20	108			2
12	1			20	77			2	30	9			30	99			3
24	1			20	138			2	38	9			20	458			2
38	1			20	293			2	50	9			20	257			2
70	1			20	86			2	80	9			40	221			4
82	1			20	299			2	19	10			20	302			2
13	2			20	459			2	23	10			20	212			2
31	2			20	551			2	25	10			20	221			2
37	2			20	381			2	33	10			20	359			2
47	2			20	336			2	35	10			30	189			3
73	2			20	189			2	37	10			20	223			2
14	3			20	225			2	39	10			40	103			4
30	3			20	627			2	41	10			20	163			2
32	3			20	528			2	43	10			30	381			3
44	3			30	277			3	45	10			20	444			2
52	3			20	468			2	47	10			30	114			3
58	3			30	255			3	49	10			20	589			2
60	3			20	191			2	57	10			20	269			2
68	3			20	547			2	71	10			40	130			4
72	3			30	325			3	73	10			20	558			2
17	4			20	133			2	75	10			20	446			2
25	4			20	294			2	10	11			20	245			2
33	4			20	248			2	14	11			20	229			2
43	4			20	491			2	34	11			20	293			2
65	4			20	284			2	36	11			20	305			2
39	6			20	163			2	38	11			30	294			3
51	6			20	282			2	40	11			50	292			5
53	6			20	317			2	42	11			20	343			2
73	6			20	154			2	44	11			20	294			2
79	6			20	155			2	46	11			20	317			2
4	7			20	356			2	48	11			30	286			3
44	7			20	205			2	54	11			20	617			2
50	7			20	197			2	76	11			30	583			3
60	7			20	635			2	78	11			40	639			4
70	7			30	635			3	33	12			30	308			3
78	7			20	662	x		2	35	12			40	308			4
7	8			30	684			3	37	12			40	324			4
11	8			20	628			2	41	12			20	309			2
19	8			20	376			2	43	12			20	266			2
35	8			20				2	45	12			30	270			3
55	8			20	305			2	47	12			40	272			4
73	8			20	162			2	49	12			30	297			3
4	9			20	160			2	51	12			30	290			3
10	9			20	261			2	53	12			20	424			2

Date: 24.11.-4713
Object: 2E-3012
Bundle: Bottom 2E-3012



Tube coordinates		biggest inner defect		biggest outer defect		biggest other defect		total def.	Tube coordinates		biggest inner defect		biggest outer defect		biggest other defect		total def.
X	Y	%	cm	%	cm	Symb.	cm		X	Y	%	cm	%	cm	Symb.	cm	
2	1			30	235			3	56	11			20	198			2
8	1			20	360			2	60	11			30	361			3
22	1			20	593			2	23	12			20	182			2
30	1			20	296			2	37	12			20	286			2
38	1			30	632			3	41	12			30	193			3
54	1			20	271			2	38	13			20	360			2
56	1			30	174			3	62	13			20	382			2
60	1			20	386			2	27	14			20	401			2
66	1			20	497			2	33	14			20	423			2
70	1			20	354			2	39	14					P		P
72	1			20	450			2	45	14			20	335			2
82	1			20	610			2	69	14			20	360			2
75	2			30	356			3	73	14			40	607			4
28	3			20	386			2	36	15			50	136			5
52	3			30	162			3	38	15			30	140			3
60	3			20	262			2	46	15			40	136			4
66	3			20	171			2	50	15			20	581			2
74	3			20	155			2									
33	4			20	195			2									
47	4			20	124			2									
77	4			20	341			2									
74	5			20	145			2									
19	6			20	434			2									
47	6			20	657			2									
4	7			20	369			2									
24	7			20	235			2									
38	7			20	234			2									
60	7			50	38			5									
47	8			30	232			3									
73	8			20	650			2									
18	9			20	670			2									
20	9			20	533			2									
36	9			20	203			2									
40	9					R	725	R									
50	9			40	224			4									
60	9			30	131			3									
66	9			40	96			4									
78	9			20	158			2									
80	9			30	124			3									
47	10			20	169			2									
32	11			20	125			2									
36	11			40	421			4									
42	11			20	254			2									
46	11			20	143			2									